Recurrence risk after a first remote symptomatic unprovoked seizure in childhood: a prospective study

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The aim of this study was to assess recurrence risk after a first remote symptomatic unprovoked seizure in childhood. All consecutive patients younger than 14 years with a first remote symptomatic unprovoked seizure who were seen at our hospital between 1994 and 2006 were included in the study and prospectively followed. Only two patients received antiepileptic treatment. Sixty-three children were included, with 35 males and 28 females. Mean age at first seizure was 4 years (SD 3y 5mo). Kaplan–Meier estimate of recurrence risk was 59% (95% confidence interval [CI] 47–71), 76% (95% CI 65-87), 85% (95% CI 76–94), and 87% (95% CI 78–96) at 6, 12, 18, and 24 months respectively. A total of 55 children out of 63 were affected by a static encephalopathy of pre- or perinatal origin. In this subgroup, recurrence risk at 12 and 24 months was 79% (95% CI 68–90) and 89% (95% CI 80–98). Univariable analysis using the Cox proportional hazards model showed that presence of global developmental delay/intellectual disability and Todd's paresis were associated with a significant increase in recurrence risk. In multivariable analysis, only Todd's paresis was significantly associated. Recurrence risk after a first remote symptomatic unprovoked seizure in childhood is much higher than what some previous studies suggests.

Recurrence risk after a first idiopathic/cryptogenic unprovoked seizure in childhood usually ranges from 32 to 50% at 2 years.[1–3] Conversely, recurrence risk after a first remote symptomatic seizure varies in different series between 57 and 96% at 2 years.[1–3] As recurrence risk is an important factor to consider when deciding about the convenience of initiating an antiepileptic treatment, the objective of the present study was to quantify more accurately recurrence risk after a first remote symptomatic unprovoked seizure in childhood.

METHOD

Cohort selection

Torrecárdenas Hospital is the reference hospital of the province of Almería, Spain. The only electroencephalogram (EEG) laboratory and paediatric neurology unit in the province are located in this hospital.

All consecutive patients aged 14 years or less who attended our unit for a first remote symptomatic unprovoked seizure between 1 June 1994 and 1 February 2006 were included in a prospective study. Neonatal convulsions, patients already on antiepileptic treatment, those who had been examined previously in other centres, and those whose detailed clinical record revealed antecedents of prior seizures were excluded. All patients were directly referred by primary care paediatricians or were first seen in the emergency department of our hospital.

All possible cases were examined by two of the authors (JRL and JAR) and only those in whom both agreed on a diagnosis of definite epileptic seizure were included. Breath-holding spells, syncope, other paroxysmal non-epileptic events, and doubtful cases were excluded.

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Definitions and classification criteria

Seizures were considered unprovoked when they occurred in absence of any known proximate precipitant. Unprovoked seizures were classified as remote symptomatic when they occurred in a participant with a history of static encephalopathy of pre- or perinatal origin or prior neurological insult such as central nervous system (CNS) infection, stroke, or significant head trauma.[4] Classification was carried out with the data available at time of first seizure. Information collected afterwards, such as a new magnetic resonance image (MRI) or neuropsychological testing, was not used. More than one seizure within a period of 24 hours was considered to be a single seizure (multiple seizure). Status epilepticus was defined as a seizure lasting more than 30 minutes or recurrent seizures lasting a total of more than 30 minutes without the patient fully regaining consciousness. Types of seizure were classified according to the 1981 International League Agand Epilepsy criteria.[5] EEGs were classified as normal or abnormal; the latter category included both epileptiform (focal or generalized spike-waves) and nonepileptiform (focal or generalized slowing) abnormalities. A family history of unprovoked seizures was defined as seizures occurring in first-degree relatives. Global developmental delay was defined as a developmental quotient <70% and intellectual disability as an intelligence quotient <70.

Initial evaluation and patient follow-up

Patients were examined within a maximum interval of 15 days by one of the participant paediatric neurologists. Family history and medical records were obtained, physical and neurological examination was performed, and an EEG and a cerebral MRI were obtained for every patient. Only two children received antiepileptic treatment after the first epileptic seizure, according to the preference of their parents. Parents were provided with a dose of rectal diazepam to use if a seizure lasted for more than 5 minutes. The study was approved by the ethics committee of the hospital. Informed consent to participate in the study was obtained from parents.

Analysis

Recurrence risk after a first remote symptomatic unprovoked seizure was calculated using Kaplan–Meier survival curves. Patients entered the study on the date of first seizure. The event under study was seizure recurrence. Cases were considered censored if, at the end of the study, the event under observation had not occurred or the patient was lost to follow-up. We carried out univariable and multivariable analyses of potential predictors of recurrence risk using the Cox proportional hazards model. We did not validate the Cox regression models using the bootstrap method since we felt that the cost in computing time was not justified for an exploratory analysis. Calculations were performed by means of SPSS for Windows, version 13.0, statistical software. The level of statistical significance was established at $p<0.05$.

RESULTS

The study group comprised 63 patients, 35 males and 28 females. Mean age at first seizure was 4 years (SD 3y 5mo). Age distribution showed predominance in the first years of life: 41% were less than 3 years and 65% less than 6 years. Mean follow-up was 4 years 1 month, range 1 month to 12 years, (SD 2y 10mo). Only one patient was lost to follow-up. In 57 patients (91%), follow-up was longer than 1 year and in 47 (75%), longer than 2 years.

General features of the cohort are shown in Table I, the aetiology in Table II, and findings on cerebral MRI in Table III. A total of 55 patients were considered affected by a static encephalopathy of pre- or perinatal origin. In addition, 23 patients were classified as unspecific developmental delay/intellectual disability, i.e. with normal MRI, no motor disability, and no definite cause.

Global recurrence risk

A total of 55 out of 63 patients relapsed, including the tool who received antiepileptic treatment. The Kaplan–Meier estimate of the recurrence risk was 59% (95% CI 47–71), 76% (95% CI 65–87), 85% (95% CI 76–94), and 87% (95% CI 78–96) at 6, 12, 18, and 24 months respectively (Fig. 1).

Table I: General features of (n=63) sample

	n	%
Prior febrile seizures	13	21
Family history of unprovoked seizures	6	9.5
Seizure type		
Partial	43	68
Generalized	20	32
Status epilepticus	5	8
Multiple seizure	17	27
Todd's paresis	5	8
Abnormal electroencephalogram	17	27
Abnormal MRI	33	52
Cerebral palsy	18	29
Global developmental delay/intellectual disability	50	79

MRI, magnetic resonance image.

Table II: Aetiology of epileptic seizures.

Group	*n*
1. Cerebral palsy ± intellectual disability	18
Spastic diplegia	11
Spastic hemiplegia	5
Spastic tetraplegia	2
2. Intellectual disability without motor disability	35
Chromosomal abnormality	3
Neonatal hypoxic–ischemic encephalopathy	4
Corpus callosum agenesis	2
Head trauma	1
Angelman syndrome	1
Sturge–Weber syndrome	1
Unspecific[a]	23
3. Without intellectual and motor disability	10
Brain tumour	2
Stroke	3
Bacterial meningitis	3
Head trauma	1
Neonatal hypoxic–ischemic encephalopathy	1

[a]With normal magnetic resonance image and no definite cause. The group of static encephalopathy of pre- or perinatal origin (55 patients) includes all patients in group 1, all patients in group 2 except the one with head trauma, and three patients in group 3: one with neonatal hypoxic–ischemic encephalopathy and two with the bacterial meningitis which occured in the first 2 weeks of life.

Table III: Findings on MRI in study group

MRI findings	*N*
Periventricular leukomalacia	11
Focal or diffuse multicystic encephalomalacia	6
Unilateral lateral ventricle enlargement	1
Parasagittal necrosis	1
Other destructive lesions[a]	8
Holoprosencephaly	1
Schizencephaly	1
Hemimegalencephaly	1
Corpus callosum agenesis	2
Typical findings of Sturge–Weber syndrome	1
Normal	30

[a]Sequelae of head trauma, bacterial meningitis, stroke, and surgery for brain tumour. MRI, magnetic resonance imaging.

In the 18 patients with cerebral palsy (CP) with or without intellectual disability, recurrence risk was 67% (95% CI 41–93) and 83% (95% CI 62–100) at 12 and 24 months respectively. Among the 23 patients with unspecific global developmental delay/intellectual disability, recurrence risk was 84% (95% CI 60–100) and 95% (95% CI 84–100) at 12 and 24 months respectively.

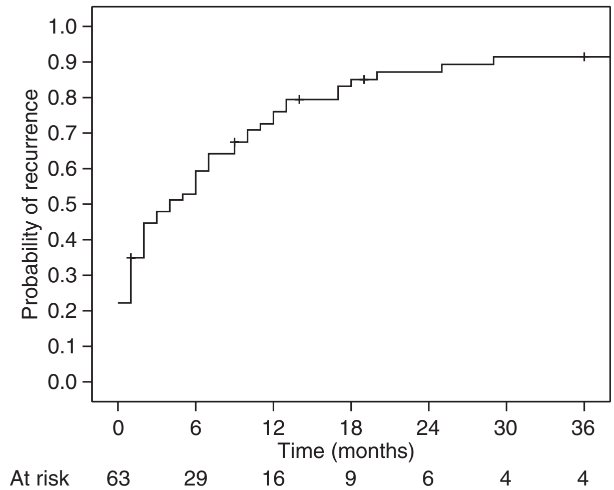

Figure 1: Recurrence risk after a first unprovoked remote symptomatic epileptic seizure in childhood. Kaplan–Meier curve.

In the 55 patients with static encephalopathy of pre- or perinatal origin, recurrence risk at 12 and 24 months was 79% (95% CI 68–90) and 89% (95% CI 80–98) respectively.

Predictors of recurrence

In order to identify potential predictors of recurrence risk we first carried out a univariable analysis using the Cox proportional hazards model. The analysis showed that global developmental delay/intellectual disability and Todd's paresis were associated with a significant increase in recurrent risk at $p<0.05$ (Table IV). Due to the small size of the sample, we decided to reduce the variables included in the multivariable analysis. Abnormal MRI, CP, global developmental delay/intellectual disability, CNS malformation, and unspecific developmental delay/intellectual disability are obviously related variables. From them, we selected the two variables with $p<0.1$: CNS malformation and global developmental dalay/intellectual disability. From the rest, we also selected the two variables with $p<0.1$: multiple seizures and Todd's paresis. Finally, prior febrile convulsions and age <3 years were included, these being the only factors associated with an increase in recurrence risk in other studies. In the multivariable analysis including these six variables, only Todd's paresis was significant at $p<0.05$. However, CNS malformation, multiple seizures, and global developmental delay/intellectual disability were associated with an increase in recurrence risk with $p<0.2$ (Table IV).

Table IV: Univariable and multivariable analysis of predictors of the recurrence risk after a first unprovoked remote symptomatic seizure; Cox proportional hazard model.

	Univariable			Multivariable		
	HR	95% CI	*p*	RR	95% CI	*p*
Prior febrile convulsions	1.5	0.8–3.0	0.190	1.2	0.6–2.4	0.638
Family history of unprovoked seizures	0.9	0.3–2.2	0.780	–	–	–
Age <3y	1.3	0.8–2.3	0.251	0.8	0.4–1.4	0.434
Multiple seizure	1.8	1.0–3.1	0.057	1.6	0.8–3.0	0.166
Status epilepticus	0.9	0.3–2.4	0.820	–	–	–
Todd's paresis	3.0	1.1–7.7	0.025	3.5	1.3–9.5	0.014
Focal seizure	0.8	0.5–1.5	0.590	–	–	–
Seizure while asleep	1.1	0.6–2.0	0.640	–	–	–
Abnormal electroencephalogram	0.7	0.4–1.2	0.180	–	–	–
Abnormal MRI	0.7	0.4–1.2	0.260	–	–	–
Cerebral palsy	1.0	0.5–1.8	0.983	–	–	–
Global developmental delay/intellectual disability	2.1	1.1–4.3	0.045	1.7	0.8–3.7	0.148
Unspecific global developmental delay/intellectual disability	1.5	0.9–2.7	0.114	–	–	–
CNS malformation	2.2	0.9–5.5	0.075	2.2	0.9–5.5	0.100
Static encephalopathy of pre- or perinatal origin	1.8	0.8–4.3	0.170	–	–	–

HR, hazard ratio; RR, recurrence risk; CI, 95% confidence interval; CNS, central nervous system.

DISCUSSION

The main limitation of our study is that our sample is hospital based. Nevertheless, it has been designed to obtain a sample as representative of general population as possible. Most patients in Almería who have epileptic seizures are cared for in Torrecárdenas Hospital since it has the only paediatric neurology unit in the province. Patients previously examined in other centres were excluded to avoid selection bias. Our sample was based exclusively on direct referrals by primary care paediatricians and patients first seen at the emergency department.

It is also important to note that only two patients in our cohort received antiepileptic treatment after the first seizure. Since both patients had recurrent seizures, treatment influence on the recurrence risk obtained in our study can be considered negligible.

Up to now, the general policy in our institution has been not to initiate an antiepileptic treatment after a first seizure, even when remote symptomatic seizures occur because recurrence risk found in previous studies, although higher than in idiopathic/cryptogenic seizures, proved to be highly variable.

In a meta-analysis published in 1991, the pooled recurrence risk after a first unprovoked seizure was 42% at 2 years.[6] Recurrence risk was higher in studies including only children (54%) than in those including only adults (43%). Since then, several prospective studies in children have shown a recurrence risk between 37 to 57% at 2 years.[1–3] Most studies reporting recurrence risk related to the presence of some kind of neurological abnormality have found increases in recurrence risk associated with such abnormalities[1,7–14] In the aforementioned meta-analysis, including studies with children and adults, recurrence risk at 2 years was 32% for patients with idiopathic/cryptogenic seizures and 57% for those with remote symptomatic seizures.[6]

Nevertheless, we should take into account that the aetiology of epilepsy changes significantly with age. Population-based epidemiological studies have shown that more than 60% of patients aged less than 15 years have a static encephalopathy of pre- or perinatal origin (CP or intellectual disability mostly), a higher percentage than at older ages.[15,16] We thus reviewed the findings in studies including only children and those concerning recurrence risk in patients with static encephalopathy of pre- or perinatal origin.

In the study by Shinnar et al., recurrence risk was 57% at 2 years. It involved a sample of 65 patients younger than 18 years old.[1] In the study by Stroink et al., recurrence risk was 74%, 27 patients aged less than 17 years were enrolled.[2] In a previous study carried out by our group, which involved a different sample of 25 children younger than 14 years old, recurrence risk was 96%.[3] Conversely, recurrence risk of idiopathic/cryptogenic seizures in these studies was relatively homogeneous (32–50% at 2y).

In two population-based studies,[9,13] recurrence risk in a small group of patients (15 and 16 patients respectively) with a static encephalopathy of pre- or perinatal origin was specifically calculated, a recurrence risk of 92 and 100% was found.

Although recurrence risk is not the only factor to be considered when deciding about the convenience of initiating an antiepileptic treatment, a recurrence risk as high as shown by some of these studies may have an influence in the decision. Consequently, the objective of our study has been to quantify more accurately recurrence risk after a first remote symptomatic seizure in childhood. We found a recurrence risk of 87% at 2 years (95% CI 78–96%). Although the confidence interval is fairly wide, the lower limit, 78%, is high enough to provide parents with useful information.

Although the authors of the other series do not report the type of neurological abnormalities present in their patients, the reason for the discrepancies observed in recurrence risk may be, at least in part, a different aetiological composition of the various samples. In our cohort, 87% of patients were affected by a static encephalopathy of pre- or perinatal origin and 79% by global developmental delay/intellectual disability. In univariable analysis, global developmental delay/intellectual disability was a significant predictor of recurrence risk. Therefore, our study does not clearly demonstrate that it is the reason for the discrepancies, but suggests that it may be a factor. Future studies on recurrence after a first epileptic seizure must detail the aetiology of the cases and not simply include in a single group all patients with remote symptomatic seizures. In any case, our study presents a useful estimate in patients with static encephalopathy of pre- or perinatal origin: 89% (CI 80–95%) at 2 years. This is in accordance with the recurrence risk observed in the small group of patients included in the two aforementioned population-based studies.[9,13]

Regarding the predictors of recurrence risk, we found that in univariable analysis, global developmental delay/intellectual disability, and Todd's paresis were associated with a significant increase in recurrence risk. In multivariable analysis, only Todd's paresis was significantly associated. Perhaps Todd's paresis is a marker of a more active disease. Only two studies are available with which we can compare these data, but their findings are not in agreement. One of them showed an increase of recurrence risk in patients with a history of prior febrile seizures and in participants aged less than 3 years.[1] In the other study,[3] univariable analysis showed a reduction in recurrence risk in patients with aged between 3 and 10 years that did not remain in multivariable analysis. It is especially relevant that abnormal EEG, an important predictor of recurrence risk in idiopathic/cryptogenic seiczures,[1,3,6] did not increase recurrence risk in remote symptomatic seizures, neither in our study nor in the other two.[1,3] A larger sample would be required in order to assess more accurately predictors of recurrence risk. Nevertheless, results may not have much clinical relevance due to elevated recurrence in all patients with remote symptomatic seizures.

CONCLUSION

Our findings show a very high recurrence risk in children with a first remote symptomatic seizure, particularly in those affected by a static encephalopthy of pre or perinatal origin. This risk is much higher than in some previous studies where these patients may have been part of a group where there were a higher number of participants affected by other type of brain lesions with a lower recurrence risk.

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